

Sho Sakaue

385-216-3305 | ✉ sho.sakaue0808@gmail.com | [in Sho Sakaue](#) | [shosakaue0808](#) | [Portfolio](#)

EDUCATION

University of Utah

Bachelor of Science in Computer Science and Applied Mathematics

Salt Lake City, Utah

Aug. 2024 - Expected Dec. 2027

- **GPA:** 3.96/4.0 and Dean's List
- **Computer Science:** Machine Learning and Deep Learning, Large Language Models, Data Structures & Algorithm, Model of Computation
- **Mathematics:** Optimization, Numerical Analysis, Probability Theory & Statistics, Linear Algebra, Real Analysis, Calculus I, II, III

EXPERIENCE

Undergraduate Research Assistant

University of Utah (Under Professor Suhas Prameela, UROP awarded)

Aug. 2025 - present

Salt Lake City, Utah

- Generated training dataset for Machine Learning Interatomic Potentials (MLIPs) across defects, surfaces, and alloy configurations from DFT calculations.
- Trained MLIPs on DFT reference datasets, comparing different optimization methods, gradient-descent or SNES-based optimization to get the best-performing model.
- Validated trained MLIPs against DFT using both training targets, including energy, force, and stress, and independent physics-based metrics, including defect energies, formation energies, and relaxation benchmarks

Mathematics Tutor and Grader

University of Utah

Aug. 2025 - present

Salt Lake City, Utah

- Tutored students in Calculus I–III, Linear Algebra, and Statistics through explanations for each student.
- Helped students strengthen mathematical reasoning, prepare for exams, and build confidence in core concepts.
- Adapted strategies to each student's learning style to improve understanding and problem-solving independence.

PROJECTS

BayesBench | *numpy, scikit-learn*

June. 2026 - present

- Implemented the benchmark on objective functions, a random search baseline, and a GP-EI optimizer for limited-budget black-box optimization.
- Added experiment scripts and plotting tools to compare optimization performance across methods.
- Structured the project as a reusable Python package for future extension to more benchmarks and optimizers.

Parameter Efficient Fine-tuning of LLM | *PyTorch, Huggingface, PEFT*

Mar. 2026 - Apr. 2026

- Implemented LoRA and QLoRA from original papers, building the low-rank adaptor and quantization pipeline.
- Fine-tuned a llama-2-1B model and benchmarked LoRA and QLoRA fine-tuned models against base and fully fine-tuned models, evaluating accuracy-performance tradeoffs.
- did experiments varying LoRA rank and quantization bits, analyzing their impact on performance and efficiency.

Multiplayer Networked Game & HTTP Server | *C#, SQL*

Nov. 2025 - Dec. 2025

- Developed a distributed multiplayer game client in Blazor using the MVC architectural pattern.
- Implemented a multi-threaded TCP network controller to communicate by JSON, using locks to ensure thread safety and prevent race conditions.
- Integrated a MySQL database to store game history and player high scores.

COMPETITION/HACKATHON

MIT iQuHACK (Quantum Computer)

Feb. 2026

- Simulated 50–60 qubit quantum circuits using Qiskit to identify hidden peak bitstrings, placing 24th in BlueQubit Challenge.
- Optimized quantum circuits simulation performance by gate reordering, reducing runtime by approximately 30%.
- Developed statistical candidate-selection heuristics to infer peak from near-uniform output distributions.

Crimson Hack (University of Utah)

Oct. 2025

- Built a rocket simulation game, where players navigate from one planet to another while avoiding obstacles and debris.
- Implemented planet-specific gravity and fuel-limited movement mechanics considering physical reality.
- Designed interactive gameplay logic that allowed users to select different solar-system planets and control rocket

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, Bash, SQL(MySQL), C#
Markup & Style Languages: HTML, CSS, L^AT_EX, Markdown
Machine Learning: PyTorch, NumPy, Hugging Face, scikit-learn, SciPy, Matplotlib, Pandas
Developer Tools: Git(GitHub, GitLab), Docker, Linux, HPC/SLURM computing

AWARDS

Frank Stenger Scholarship	Fall 2026
Undergraduate Research Opportunity Program (UROP)	Summer 2026
ISSS Scholarship	Spring 2026